

RESEARCH ARTICLE



Genotype accounts for intraspecific variation in the timing and duration of multiple, sequential life-cycle events in a willow species

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Abstract

Premise: Phenological variation among individuals within populations is common and has a variety of ecological and evolutionary consequences, including forming the basis for population-level responses to environmental change. Although the timing of life-cycle events has genetic underpinnings, whether intraspecific variation in the duration of life-cycle events reflects genetic differences among individuals is poorly understood.

Methods: We used a common garden experiment with 10 genotypes of *Salix hookeriana* (coastal willow) from northern California, United States to investigate the extent to which genetic variation explains intraspecific variation in the timing and duration of multiple, sequential life-cycle events: flowering, leaf budbreak, leaf expansion, fruiting, and fall leaf coloration. We used seven clones of each genotype, for a total of 70 individual trees.

Results: Genotype affected each sequential life-cycle event independently and explained on average 62% of the variation in the timing and duration of vegetative and reproductive life-cycle events. All events were significantly heritable. A single genotype tended to be “early” or “late” across life-cycle events, but for event durations, there was no consistent response within genotypes.

Conclusions: This research demonstrates that genetic variation can be a major component underlying intraspecific variation in the timing and duration of life-cycle events. It is often assumed that the environment affects durations, but we show that genetic factors also play a role. Because the timing and duration of events are independent of one another, our results suggest that the effects of environmental change on one event will not necessarily cascade to subsequent events.

KEYWORDS

coastal willow, heritability, genetic variation, National Phenology Network, phenology, plasticity, Salicaceae, *Salix hookeriana*

The phenology of life-cycle events is important because it influences components of fitness, such as survival and reproduction (Primack, 1985; Rathcke and Lacey, 1985;

Forrest and Miller-Rushing, 2010; Semenchuk et al., 2016; Iler et al., 2021). Phenological distributions are most often described quantitatively by their timing (e.g., onset, peak, or

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end), but other important aspects of a phenological distribution include duration, synchrony, and skewness (Rathcke and Lacey, 1985). The timing and duration of vegetative and reproductive life-cycle events in plants have a variety of ecological consequences in seasonal environments, via abiotic and biotic pathways (Iler et al., 2021). The timing and duration of leaf expansion affect vegetative growth, flowering, reproductive output, and survival via both the length of the growth period and exposure to herbivores (Wipf et al., 2006; Wipf, 2010; Liu et al., 2011a; Robson et al., 2013; Jacques et al., 2015; Choi et al., 2019). Additionally, the timing and duration of flowering influence reproductive output via exposure to favorable or unfavorable abiotic conditions (e.g., frost risk; Inouye, 2008; Augspurger, 2013) and indirect interactions between plant species for pollinators by moderating the degree of co-flowering among plant species (Waser, 1978; Waser and Real, 1979; Faust and Iler, 2022). The degree of within-population variation in the phenology of life-cycle events is important to understand because it is the variation on which natural selection can act, and it therefore determines the ability of populations to respond to environmental change.

Variation in the timing of vegetative and reproductive life-cycle events among individual plants within populations is common (Fox et al., 1997; Wesołowski and Rowiński, 2006; Salmela et al., 2013; Delpierre et al., 2017; Troth et al., 2018), and the timing of events is typically considered to be highly heritable (Elzinga et al., 2007; Forrest and Miller-Rushing, 2010). Although the timing of life-cycle events clearly has genetic underpinnings (e.g., Donohue et al., 2010; Anderson et al., 2012; Troth et al., 2018; Martínez-Berdeja et al., 2020), the degree to which the duration of life-cycle events reflects genetic differences among individuals is largely unknown (Elzinga et al., 2007). In contrast to timing, the duration of life-cycle events is typically considered to be strongly influenced by abiotic and biotic components of the environment (Bock et al., 2014; Semenchuk et al., 2016). For example, temperature and pollinator activity affect the amount of time an individual plant maintains open flowers (Dunne et al., 2003; Bock et al., 2014), as well as the longevity of individual flowers, which is likely to affect flowering duration at the plant level (Primack, 1985). In ecosystems with seasonal snow cover, the timing of spring snowmelt influences the number of days over which plants maintain open flowers and ripen fruit (Jerome et al., 2021), and temperature and photoperiod determine the number of days over which plants grow (Bauerle et al., 2012). Although the duration of life-cycle events is clearly influenced by the environment, duration is also likely to be under some degree of genetic control; however, this is poorly understood (Elzinga et al., 2007).

Both the timing and duration of events early in the life cycle might affect the timing and duration of subsequent life-cycle events (Li et al., 2016; Jerome et al., 2021; Sandor et al., 2021). That is, sequential life-cycle events might not

be independent of one another. For example, if there is a fixed developmental window between two life-cycle events, the second life-cycle event might occur early relative to other individuals in the population simply because the first life-cycle event occurred early (i.e., an individual might exhibit early leaf expansion, simply because leaf budbreak was early). This potential influence of early events on later events emphasizes the importance of studying multiple life-cycle events simultaneously. Such an approach will improve our understanding of the extent to which genetic variation plays a role in influencing the timing and duration of sequential life-cycle events, or whether early events simply determine the timing and duration of later events. Additionally, it is unclear whether individuals of the same genotype that exhibit an early event relative to the population will be consistently early for other life-cycle events, or whether the order of genotypes varies across life-cycle events (and if the same is true for short vs. long duration of events).

Here we used a common garden experiment consisting of individual clones of multiple genotypes from a population of *Salix hookeriana* (coastal willow) to investigate the amount of intraspecific variation in the timing and duration of multiple, sequential life-cycle events (flowering through leaf senescence) and to examine the degree to which genetic variation contributes to such variation. We asked the following questions: (1) Does genetic variation (measured as genotype) account for intraspecific variation in the timing and duration of life-cycle events? (2) Are timing and duration heritable? (3) Does genotype affect sequential life-cycle events independently, or are later events (like senescence) affected by earlier events (like leaf expansion)? (4) Are genotypes consistently ordered from early to late (or short to long) across life-cycle events, or is there variation in which genotypes are early vs. late, depending on the life-cycle event?

MATERIALS AND METHODS

Study site

This study was conducted in northern California, USA, at the Humboldt Bay National Wildlife Refuge in the year 2014 (HBNWR; 40°40'53"N, 124°12'4"W). The refuge is composed of a mix of coastal dunes, salt marsh, riparian wetlands, and forest habitat. The HBNWR climate is mediterranean with cool, moist winters and summertime temperatures that are moderated by its close proximity to the ocean. Average annual precipitation is 1170 mm, and almost all precipitation occurs as rain from October to May. For our study, we used a flat, 50 × 150 m pasture because it provided an area large enough for our common garden. The pasture used to be a salt marsh and was converted to agricultural uses approximately 120 years ago. The pasture was dominated by non-native grasses and had no woody vegetation prior to our experiment.

Study species

Salix hookeriana (coastal willow; Salicaceae) is a woody, deciduous species that grows as a small tree or shrub. It ranges from northern California to Alaska and grows in wet habitats such as coastal dunes, floodplains, and meadows (Argus, 2013). *Salix hookeriana* is dioecious and likely reproduces sexually via wind or animal pollination, in addition to asexually via vegetative growth (Sacchi and Price, 1988; Argus, 2013). Willows are useful study species for questions pertaining to genetic variation because populations often contain a high degree of genetic variation, and this genetic variation is also associated with large differences in traits within populations (Fritz and Price, 1988; Brunsfeld et al., 1991).

Experimental design

Stem cuttings (25 cm) were haphazardly collected from 27 individuals of *S. hookeriana* around Humboldt Bay in February 2009 (14 females and 13 males, 25 cuttings per individual). The area around Humboldt Bay includes numerous clusters of willow, all found in relatively homogenous low-elevation riparian and marsh habitat. Same-sex willows from which we made cuttings were at least 100 m apart except for one pair (~60 m apart). Stem cuttings were soaked in water for 2 weeks before planting in a common garden at the HBNWR (Appendix S1). The common garden was a former cattle pasture that was fenced in to prevent deer and cattle herbivory. Stems were planted 3 m apart in a randomized location within a 45 × 135 m grid within the 50 × 150 m pasture. Our observation of willows in the region suggests that the common garden planting density is similar to that found in natural settings. A total of 25 stems from each individual was planted, for a grand total of 675 willows (25 stems each from 27 individuals). Stem cuttings were genotyped using two microsatellite markers, after which 26 individuals were identified as unique genotypes (13 females, 13 males; described by Barbour et al. [2015]). Although relatedness among the genotypes is unknown, previous work shows that the trait space of the phenotypes is distinct, suggesting that we can treat the genotypes as independent units (Barbour et al., 2015).

In 2014, we selected seven clonal replicates of a random subset of 10 of the genotypes (5 female and 5 male, $N = 70$ individuals) to be surveyed for phenology every 2 weeks for a single growing season. At the time of our study, the individual willows had reached heights of 8–10 m. The phenology of five life-cycle events was recorded based on the USA National Phenology Network's monitoring scheme (USA National Phenology Network NPN, 2020), listed in order of occurrence: flowering, leaf budbreak, leaf expansion, fruiting, and leaf coloration. Flowering, leaf budbreak, and fruiting were recorded as count estimates on each individual shrub, using a defined range of values: <3, 3–10, 11–100, 101–1000, 1001–10,000. Leaf expansion and leaf

coloration were recorded as percentage values on each individual shrub, using the following categories: <5%, 5–24%, 25–49%, 50–74%, 74–95%, >95%. We only included leaf coloration that occurred in late summer/early fall, as opposed to spring. Seven individuals exhibited a small percentage of leaf coloration in the spring, which we assumed was due to some kind of stress, as opposed to the start of senescence (e.g., nutrient deficiency, water stress; Hopkins and Hüner, 2009). Phenology surveys occurred every 2 weeks from 15 February–9 May, at which time all spring events had concluded (flowering, leaf budbreak, leaf expansion, and fruiting) and again from 9 September–17 December for fall leaf coloration.

In 2020, the size of each individual willow was assessed on 7–11 March. We measured the diameter (cm) of the trees at 10 cm above ground using a diameter tape (Forestry Suppliers, Jackson, MS, USA). However, most trees were multitemmed at or below 10 cm, and in these cases, diameter of all stems were measured, rather than taking a single diameter that would include “empty space” between stems. For each tree, we summed the diameter measurements across all stems as a relative measure of plant size.

Analyses

All analyses were performed in R v. 4.0.2 (R Core Team, 2020). For each individual plant, we calculated two metrics from the phenology surveys: date of peak occurrence and duration (last date – first date + 1) for each event. The first date was defined as the first day on which flowers, leaf buds, expanding leaves, fruits, or colored leaves (yellow leaves) were observed, and the last day as the last day on which any of these stages were observed. We used peak date to represent the timing of life-cycle events because peak dates are less prone to error than first and last (Miller-Rushing et al., 2008), especially considering that sampling took place every 2 weeks. Peak was defined as the day on which 50% of flowers, leaf buds, or fruits were counted, or the day on which 50% of leaves had emerged, or 50% of leaves had color. This date was determined using linear interpolation for events with count data (flowering and fruiting; Høye et al., 2007; Iler et al., 2013). We collapsed each range from the phenology surveys into a single value for analysis, using the upper value of the range. For example, a count of <3 became 3, a count of 3–10 became 10, a percentage <5% became 5%, a percentage of 5–24% became 25%, with the >95% category becoming 99%. Thus, peak date for the percentage categories (leaf expansion and coloration) was reached when anywhere from 24–49% of the leaves had expanded or colored, as opposed to 50–74%; our reasoning is that we would rather err on the side of an event approaching peak than an event being past its peak. The duration of an event was calculated as the last date minus the first date, plus one.

We first determined whether genotype affects phenological events independently, or whether earlier events affect

later events, using multivariate analysis of variance (MANOVA). Sequential life-cycle events may not be independent from one another, which could lead to dependent events changing together (Li et al., 2016; Jerome et al., 2021), as opposed to each one responding independently to genotype. We therefore conducted two separate MANOVAs, one for timing (peak dates) and one for duration, using Pillai's test statistic. Durations were log-transformed.

We then analyzed each life-cycle event on its own, using linear mixed effect models (LMMs) in the package *glmmTMB* (Brooks et al., 2017). Most responses were log-transformed to meet the assumptions of normality and heteroscedasticity of residuals. Fruiting date and leaf coloration date were not log-transformed. Planting row was included as a random intercept term to account for any potential effect of microclimatic variation across the common garden that could potentially affect phenology. For timing responses, genotype, sex, and plant size were used as predictors. For duration responses, we also included timing as a predictor, because the timing of events can influence their duration (e.g., an earlier-flowering individual may flower for more days; Ollerton and Lack, 1998). We conducted model selection for each response (timing and duration) using the package *bbmle* (Bolker, 2021). We included all predictors as additive effects, in all possible combinations, except that we did not include sex and genotype in the same model because this prevented model convergence. Models with genotype always outperform the analogous model with sex (Appendix S2, Tables S1, S2), indicating that sex does not have a large effect on the timing or duration of life-cycle events. Thus, we had six candidate models for the phenological events and 13 for the duration of events (four and eight models for timing and duration of fruiting, respectively, because only female plants produce fruit). The model with the lowest Akaike information criterion (AIC) value was selected as the best model. We report marginal R^2 values for models of best fit (i.e., pertaining to the fixed effects), calculated using package performance (Lüdtke et al., 2021).

From our LMMs, we extracted the residual sums of squares among clones, which represents the amount of variation explained within genotypes (i.e., plasticity), and then divided this by the total sums of squares to represent the proportion of intraspecific variation explained by plasticity. As this study was not designed to test a spectrum of responses to climatic variation, our measure of plasticity is likely an underestimation of total potential genotypic plasticity. It does, however, quantify the degree of plasticity present within the study population for this common garden environment.

We next calculated broad-sense heritability for each trait (H^2) as the proportion of the total phenotypic variance in each trait (V_P) explained by genotypic variance (V_G): $H^2 = V_G/V_P$ (Lynch and Walsh, 1998). H^2 ranges from 0 to 1, with values close to 1 indicating high heritability (trait is strongly affected by genetic variation) and values close to 0 indicating low heritability (trait is strongly

affected by environmental variation). To calculate genotypic variance (V_G), we fit a linear mixed effects model with genotype as a random effect and extracted the variance of the random effect from this model for each trait (using package *nlme*; Pinheiro et al., 2021). V_P is the sum of the residual and genetic variance from these models. To determine whether H^2 was significant for each trait, we fit an intercept-only model and compared this to the model with genotype as a random effect, using likelihood ratio X^2 tests (Barbour et al., 2015).

To determine whether genotypes exhibit a consistent order across life-cycle events (e.g., whether early genotypes are early for all life-cycle events), we conducted Spearman rank correlations. We ranked the order of the genotypes for each life-cycle event from earliest to latest (for phenology) and shortest to longest (for duration) based on the mean value for each event and genotype. We performed a correlation test for each pairwise comparison of events, for a total of 10 correlation tests for phenology and 10 for duration (5 life-cycle events for each result in 10 tests). If $r = 1$, then the order of the genotypes is identical between two events.

RESULTS

Effects of genotype on timing and duration

The timing of a given event occurred over a range of approximately 1 to 2 months across all individuals in our sample, depending on the specific event; for example, peak fruiting, the event with the least amount of variation, occurred over 39.7 days, whereas peak leaf coloration occurred over 70 days (Figure 1A). Genotype had a significant effect on the timing of all life-cycle events: peak flowering, peak budburst, peak leaf expansion, peak fruiting, and peak leaf color date, even after accounting for any potential effects of plant size on the timing of life-cycle events (Table 1; Figure 2). Genotype explained 49–80% of the variation in timing across individuals, depending on the life-cycle event (mean = 65%; Table 1). Plant size only affected peak leaf expansion and peak leaf coloration date, and plant size was not included in the best model for other life-cycle events (Appendix S2, Table S1). We found no evidence that sex plays an important role in explaining intraspecific variation in the timing of life-cycle events in this species, as models including sex as a predictor performed substantially worse than those including genotype (at least 37.6 AIC units below the best model, which always included genotype; Appendix S2, Table S1). Finally, plasticity within the common garden explained less variation in the timing of events than did genotype for all events except fruiting, ranging from 20–51% (mean = 36%; Table 1).

There was a similar amount of intraspecific variation in the duration of life-cycle events, with durations varying by 38 days across individuals (for the duration of fruiting) and up

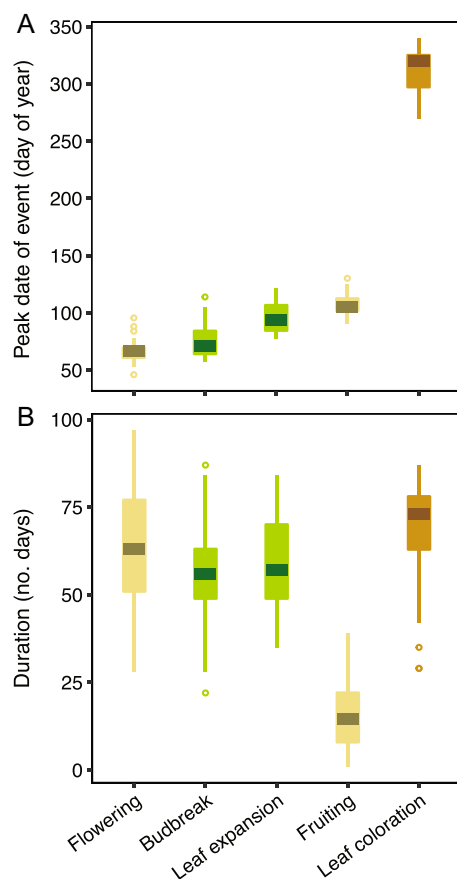


FIGURE 1 Intraspecific variation in the (A) timing and (B) duration of life-cycle events in *Salix hookeriana* in a common garden in California. Box plots represent all 70 individuals in the study, pooled across genotype. Day of year is shown as number of days from 1 January.

to 69 days (duration of flowering; Figure 1B). Genotype also had a significant effect on the duration of all life-cycle events: peak flowering, peak budburst, peak leaf expansion, peak fruiting, and peak leaf color date, even after accounting for potential effects of plant size and phenology on the duration of life cycle events (Table 2, Figure 3). Genotype explained 39–82% of the variation in phenology across individuals, depending on the life-cycle event (mean = 62%; Table 2). Phenology only affected the duration of leaf expansion, with later plants exhibiting longer durations, and phenology was not included in the best model for other life cycle events (Appendix S2, Table S2). Plant size was not included in the best model for any life-cycle event (Appendix S2, Table S2). We found no evidence that sex plays an important role in explaining intraspecific variation in the duration of life-cycle events in this species because models including sex as a predictor performed substantially worse than those including genotype (at least 19.3 AIC units below the best model, which always included genotype; Appendix S2, Table S2). Finally, plasticity explained less variation in the duration of events than did genotype for all events except leaf budbreak, ranging from 16 to 48% (mean = 35%; Table 2).

Heritability

Consistent with model selection results that genotype explains an important amount of intraspecific variation in timing and duration, timing and duration were significantly heritable (Table 3). Furthermore, most events had H^2 values above 0.50 (except for the duration of leaf budburst, fruiting, and leaf coloration).

Are sequential events independent?

The peak date of life-cycle events in *Salix hookeriana* each responded independently to genotype (MANOVA: $Pillai = 1.55$, approximate $F_{20,72} = 2.28$, $P = 0.0058$). Similarly, the log-transformed duration of life-cycle events also responded to genotype independently from one another ($Pillai = 1.88$, approximate $F_{20,72} = 3.19$, $P = 0.0002$).

Order of genotypes across events

Genotypes maintained a fairly consistent rank order for the timing of events but not for the duration of events (Table 4). For timing, all the significant correlations were in the positive direction, indicating that early genotypes tend to be early and late genotypes tend to be late. This result mostly held for every event except for leaf coloring. For fruiting date and flowering date, the rank orders of genotypes were perfectly correlated ($r = 1.0$; indicating that the rank order of genotypes was identical between the two life-cycle events); although the rank orders of genotypes were not significantly correlated for budbreak and fruiting or leaf expansion and fruiting, they were both correlated with fruiting date ($r = 0.70$, $P = 0.23$ and $r = 0.90$, $P = 0.083$, respectively). In contrast, for the duration of life-cycle events, only one pair of events exhibited a significant correlation in the rank order of genotypes (leaf budburst and leaf expansion duration; Table 4; all other P -values > 0.11).

DISCUSSION

Our common garden experiment showed that genetic variation can be a major component underlying within-population variation in the timing and duration of vegetative and reproductive life-cycle events. *Salix hookeriana* individuals of the same genotype were more likely to conduct life-cycle events at the same time and for a similar duration compared to individuals of different genotypes. Genotype explained intraspecific variation in phenological traits even after accounting for potential effects of earlier events on later events. Our experiment also revealed the presence of “early” vs. “late” genotypes; in contrast, the duration of events could not be classified as “short” vs. “long” because a single genotype exhibited a combination of short vs. long event durations.

TABLE 1 Effects of genotype and plant size on intraspecific variation in the timing of life-cycle events in *Salix hookeriana* (coastal willow) in a common garden in northern California. Predictors are from LMMs with planting row as a random intercept term and peak date of each life cycle event as a response. Only the results from the best model (lowest AIC value) are shown here. Candidate set of models can be found in Appendix S2, Table S2. R^2_m refers to the percentage variation explained by the fixed effects of the model, which typically only contain genotype. R^2_c refers to the percentage variation explained by the fixed and random effects; the only random effect in our models was planting row.

Life cycle event	Predictor	X^2	df	P	R^2_m	$R^2_{\text{plasticity}}$	R^2_c
Flowering date	Genotype	116.4	9	<0.0001	0.63	0.35	0.64
Leaf budbreak date	Genotype	293.5	9	<0.0001	0.80	0.20	0.82
Leaf expansion date	Genotype	136.8	9	<0.0001	0.72	0.28	0.72
	Size	16.3	1	<0.0001			
Fruiting date	Genotype	21.0	3	0.0001	0.49	0.51	0.49
Leaf coloration date	Genotype	83.83	9	<0.0001	0.55	0.45	0.59

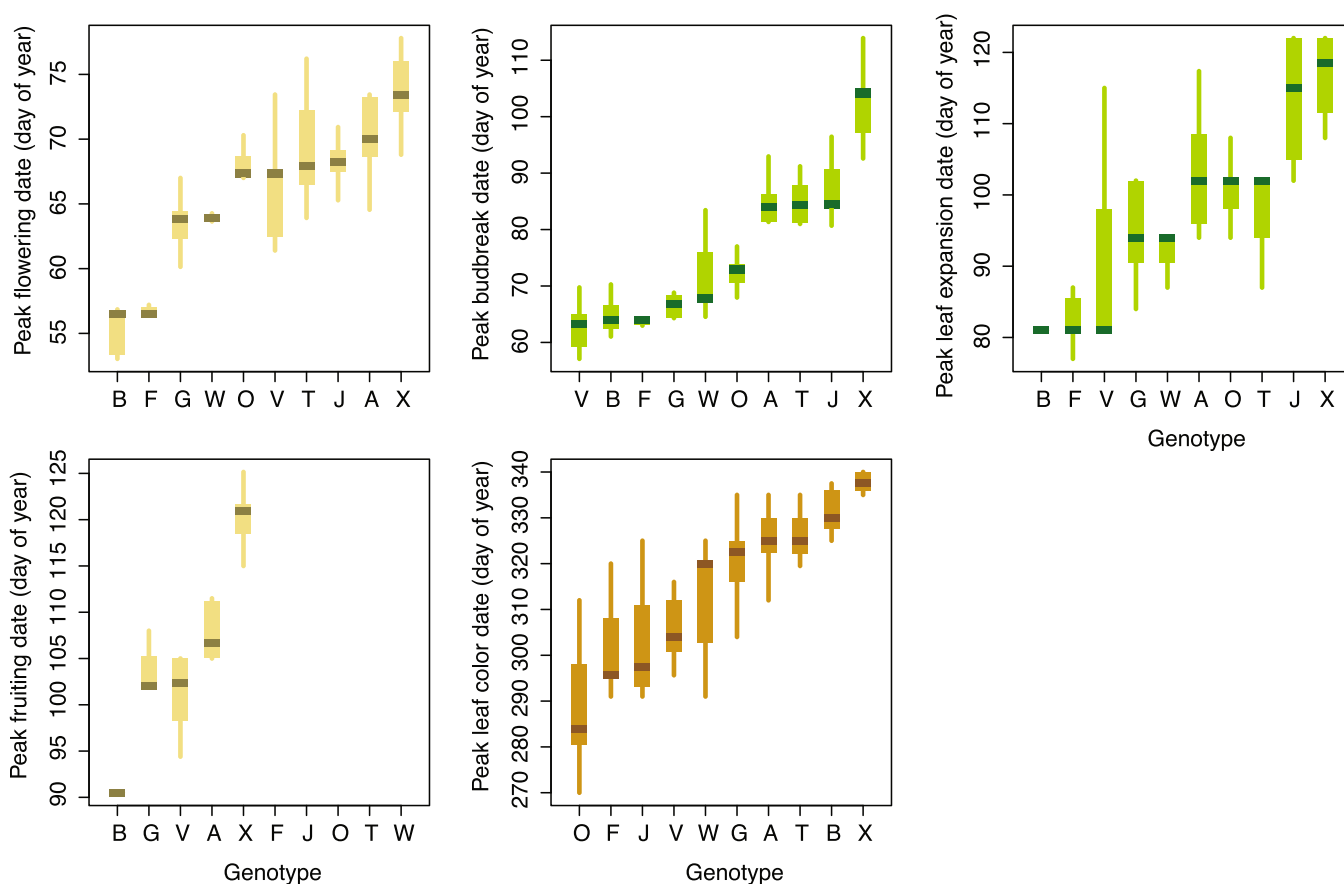


FIGURE 2 The effect of genotype on the phenology of life-cycle events in *Salix hookeriana* in a common garden in California. Box plots represent seven genetically identical individuals for each genotype. Day of year is shown as number of days from 1 January. Straw color represents reproductive events, green leaf out events, and tan leaf coloration.

The intraspecific variation in the timing and the duration of life-cycle events within our study population is striking. The magnitude of intraspecific variation in the timing of single events is comparable to the magnitude of interspecific variation in leaf-out dates in botanic gardens, including from samples of up to 444 tree species (Panchen et al., 2014). Our study provides important insight into the degree of

intraspecific variation in the duration of life-cycle events, which to our knowledge has not been shown to this extent. However, there is room for error in our calculation of event duration, with start and end dates potentially occurring earlier than observed, given that phenology was censused every 2 weeks. This should not result in any directional bias in terms of duration being under- or overestimated. In cases

TABLE 2 Effects of genotype and plant size on intraspecific variation in the duration of life-cycle events in *Salix hookeriana* (coastal willow) in a common garden in northern California. Predictors are from LMMs with planting row as a random intercept term and peak date of each life cycle event as a response. Only the results from the best model (lowest AIC value) are shown here. Candidate set of models can be found in Appendix S2, Table S2. R^2_m refers to the percentage variation explained by the fixed effects of the model, which typically only contain genotype. R^2_c refers to the percentage variation explained by the fixed and random effects; the only random effect in our models was planting row.

Life cycle event	Predictor	X^2	df	P	R^2_m	$R^2_{\text{plasticity}}$	R^2_c
Flowering duration	Genotype	351.7	9	<0.0001	0.82	0.16	0.86
Leaf budbreak duration	Genotype	64.5	9	<0.0001	0.39	0.48	0.63
Leaf expansion duration	Genotype	134.3	9	<0.0001	0.67	0.32	0.72
	Phenology	5.7	1	0.017			
Fruiting duration	Genotype	24.7	3	<0.0001	0.65	0.36	0.65
Leaf coloration duration	Genotype	84.8	9	<0.0001	0.55	0.44	0.56

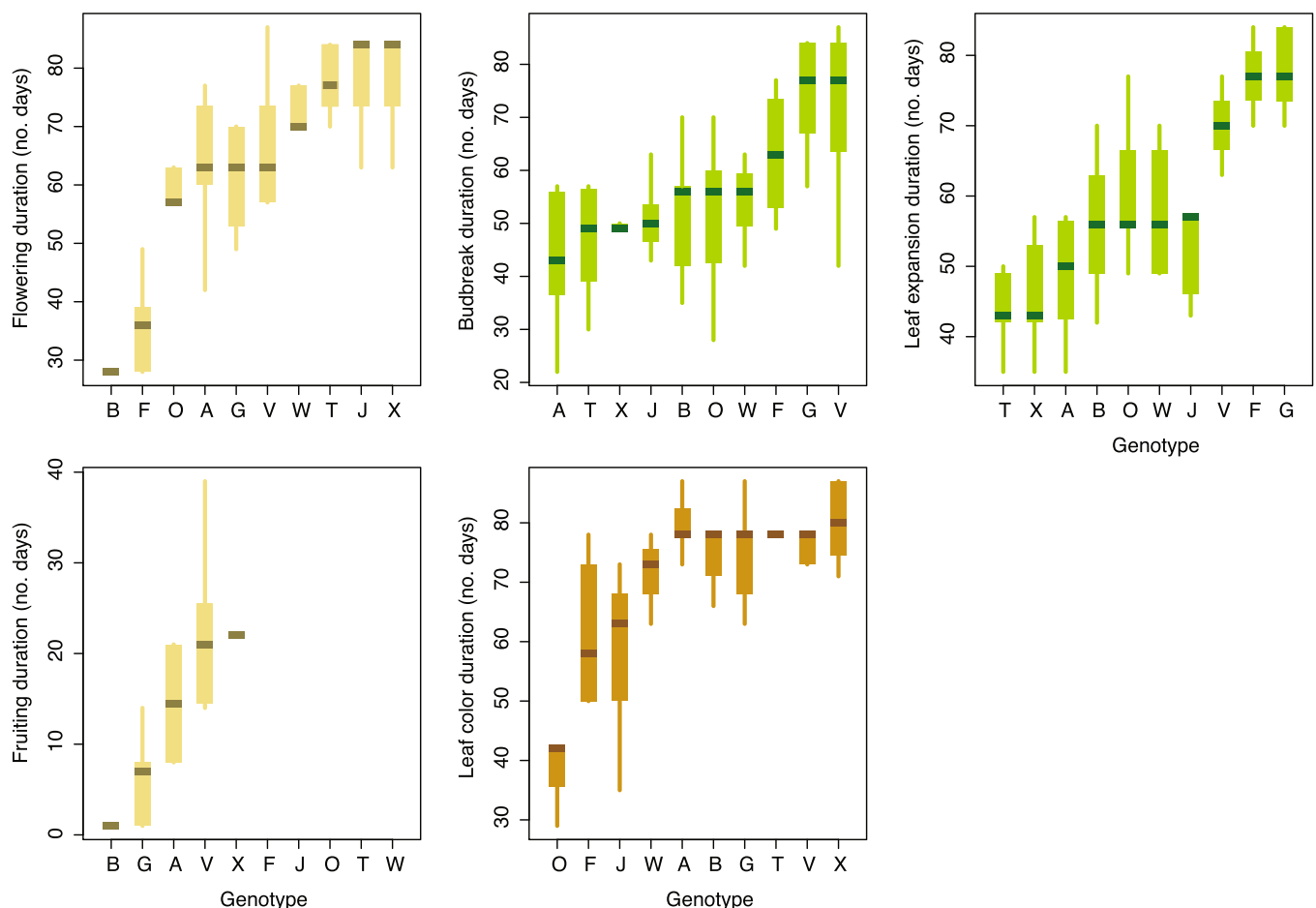


FIGURE 3 The effect of genotype on the duration of life-cycle events in *Salix hookeriana* in a common garden in California. Box plots represent seven genetically identical individuals for each genotype. Straw color represents reproductive events, green leaf out events, and tan leaf coloration.

in which durations were overestimated, it would be by at most 13 days, making our largest observed range for the duration of an event (70 days) from anywhere between 57–83 days, which is still quite variable. Substantial phenotypic diversity is often observed in populations of forest trees, including for phenological traits (Scotti et al., 2016). Also

consistent with our findings, other *S. hookeriana* traits are highly variable across individuals and genotypes within this common garden (Barbour et al., 2015, 2016).

In our common garden experiment, much of the within-population variation in phenological traits was explained by genotype: 49–80% for timing and 39–82% for duration,

depending on the event. This result is consistent with populations of *Salix* spp. in general exhibiting a high degree of genetic and phenotypic variation (Fritz and Price, 1988; Nichols-Orians et al., 1993), which likely results from the presence of multiple selective agents on a variety of traits (Barbour et al., 2015, 2016). Additionally, we found that most events are moderately to highly heritable, with only the duration of leaf budburst and leaf coloration exhibiting relatively low heritability estimates. One potential benefit of this phenological variation and its genetic underpinnings is that the risk of suboptimal environmental conditions (abiotic or biotic; Wolkovich et al., 2014) from flowering or leafing out too early or too late (or for too short or too

long) is distributed across a range of dates and therefore individuals, creating opportunity for selection on both timing and duration of life-cycle events. Therefore, *S. hookeriana* in our study area has the capacity to be resilient to environmental change. At the same time, multiple selective agents might constrain the extent to which timing and duration can adapt to any particular environmental change or factor (Brody, 1997). Although broad-sense heritability accounts for additive, dominant, and epistatic variation, research on a variety of complex traits across a diversity of species suggests that most genetic variance is likely additive (Hill et al., 2008). Thus, most of the phenological variation in our study is likely to represent additive genetic variation on which natural selection can act.

The duration of life-cycle events is usually hypothesized as being influenced by abiotic and biotic components of the environment (Bock et al., 2014; Semenchuk et al., 2016), but here we showed that event durations can be strongly influenced by genetic variation and can be highly heritable. Indeed, research at the scale of both individual plants and populations shows that the duration of flowering, leaf budbreak, leaf expansion, and fruiting is influenced by temperature, precipitation, snowmelt timing, light, and photoperiod (Seiwa, 1999; Dunne et al., 2003; Bock et al., 2014; Gömöry et al., 2015; Jerome et al., 2021; Li et al., 2021). Thus, for *S. hookeriana*, genetic constraints might limit the extent to which the abiotic and biotic environment affects event durations. Our results are consistent with a study showing that the duration of leaf and flowering events in several Arctic plant species appears to be fixed, suggesting that event durations have a genetic basis (Semenchuk et al., 2016). In contrast, it is often hypothesized that the timing of an event will affect its duration (i.e., durations are not fixed)—for example, earlier-flowering individuals tend to flower for more days (e.g., Ollerton and Lack, 1998; Semenchuk et al., 2016), but we generally did not find evidence to support this

TABLE 3 Broad-sense heritability (H^2) for phenological and duration traits. H^2 ranges from 0 to 1, with values close to 1 indicating high heritability (trait is strongly affected by genetic variation) and values close to 0 indicating low heritability (strongly affected by environmental variation). Likelihood ratio X^2 tests compare a model with genotype as a random effect to a model without the random effect, as a test of whether heritability is significant.

Trait	H^2	X^2	P
Flowering date	0.56	32.05	<0.0001
Budburst date	0.78	69.99	<0.0001
Leaf expansion date	0.61	39.25	<0.0001
Fruiting date	0.55	5.90	0.015
Leaf coloring date	0.50	25.98	<0.0001
Flowering duration	0.74	60.12	<0.0001
Budburst duration	0.40	17.28	<0.0001
Leaf expansion duration	0.65	45.15	<0.0001
Fruiting duration	0.49	5.33	0.021
Leaf coloring duration	0.20	5.09	0.024

TABLE 4 Correlation matrices of the rank order of the timing and duration of life-cycle events among genotypes, from Spearman rank correlation tests for each pair of life-cycle events (separately for timing and duration). Significant correlations are shown in bold (at $P < 0.05$). * $P = 0.083$. All other P -values are > 0.11 .

Event type	Event	Budburst	Leaf expansion	Fruiting	Leaf coloring
Timing	Flowering	0.87	0.90	1.0	0.30
	Budburst	—	0.90	0.70	0.32
	Leaf expansion	—	—	0.90*	0.15
	Fruiting	—	—	—	0.30
	Leaf coloring	—	—	—	—
Duration	Flowering	−0.33	−0.53	0.80	0.19
	Budburst	—	0.93	−0.10	−0.40
	Leaf expansion	—	—	−0.20	−0.48
	Fruiting	—	—	—	0.60
	Leaf coloring	—	—	—	—

hypothesis. The exception was that earlier individuals had a shorter period of leaf expansion; this relationship might relate to herbivory because expanding leaves are more vulnerable to herbivory compared to mature leaves or leaf buds (Moles and Westoby, 2000; Liu et al., 2011b). Therefore, when herbivory varies seasonally, there could be a benefit to expanding leaves early and quickly to limit herbivore damage, but this hypothesis requires testing. Finally, the effects of genotype on duration in our study occurred independently of plant size, in contrast to other work showing that larger plants have longer flowering durations (Ollerton and Lack, 1998). Plant size might not have affected event durations in our study because all individuals were planted at the same time; there would likely be more size variation within a natural population.

We demonstrated that early genotypes tend to be early, and late genotypes tend to be late, at least for all four spring life-cycle events: the timing of flowering, leaf budbreak, leaf expansion, and fruiting. This result is not driven by sequential events affecting one another. Instead, our results show that there is a genetic basis for the timing of *each* trait. Because the timing and duration of life-cycle events were independent of one another, our results suggest that the effects of environmental change on one event will not necessarily cascade to subsequent events. This pattern did not hold for the timing of fall leaf senescence, perhaps because this event occurred at such a different time in the life cycle and might therefore reflect different underlying genetic mechanisms. Although the rank order of genotypes was strongly correlated for fruiting and leaf budbreak and for fruiting and leaf expansion ($r = 0.70$ and $r = 0.90$, respectively), these were either not statistically significant or were marginally significant ($P = 0.23, 0.083$, respectively). This lack of statistical significance is likely due to the small sample size for fruiting, where we had five instead of 10 genotypes. Yet these correlations, especially with such a small sample, suggest a biologically meaningful relationship. Our results are consistent with a study of five woody species in which individuals exhibited a consistent order of leaf out from one year to the next, suggesting genetic differences between early vs. late individuals (Wesołowski and Rowiński, 2006). We observed no evidence for trade-offs (negative correlations), in which the timing or duration of one event constrains the timing or duration of another event. In contrast to the timing of life-cycle events, genotypes exhibit a combination of short vs. long durations of events (e.g., short flowering and long leaf expansion). This result suggests a role for environmental variation, in addition to genetic variation, in affecting the duration of life-cycle events, as discussed above. There was, however, one strong and significant correlation in the rank order of genotypes between the duration of leaf budbreak and leaf expansion: Genotypes with short budbreak tended to also have short leaf expansion periods (vice versa for long periods). Future work should investigate whether genotypes tend to be early or late (or short or long) across multiple life-cycle events in other species, to determine whether

generalizations can be made about the phenology of unmeasured life-cycle events based on measured ones.

Although we showed that most variation in phenology is explained by genotypic differences, we additionally found substantial variation in timing and duration among clones of the same genotype, which reflects trait plasticity (Kramer, 1995; Vitasse et al., 2010). In other words, a single genotype in the same site expressed different phenotypes, which was likely a response to micro-environmental variation across the common garden. It is unclear what is driving this variation in phenological traits among clones in our experiment, but likely temperature, soil moisture, and/or some interaction of these with photoperiod are likely to play a role, given that these environmental factors are known to cue vegetative and reproductive life-cycle events in plants (Rathcke and Lacey, 1985; Forrest and Miller-Rushing, 2010; Bauerle et al., 2012). For example, in *Quercus robur*, *Quercus petraea*, and *Fagus sylvatica*, some of the variation in the timing of leaf unfolding and leaf senescence among individual trees appears to be caused by fine-scale variation in access to soil moisture (Delpierre et al., 2017). Because there is some overlap of willow leaf canopies across the common garden, it is also possible that interactions among individual trees might influence light availability and shading, and therefore the timing and duration of events (Seiwa, 1999; Augspurger and Bartlett, 2003). As previously mentioned, our experiment was not designed to quantify induced plasticity in response to climatic variation; thus, our measure of plasticity is likely an underestimation of what is present within the study population. Although it is unclear whether environmental cues of phenology generally vary sufficiently within populations to account for observed magnitudes of intraspecific phenological variation (Gressler et al., 2015), our study suggests that they might vary sufficiently even over a relatively homogenous area to account for some within-population variation in phenological traits. However, much more research on this topic is needed.

Although the fitness implications of the timing and duration of each genotype for *S. hookeriana* are unknown, we explore several hypotheses to guide future research. Genotypes that break leaf buds and expand their leaves earlier might have a competitive growth advantage, or they might experience different magnitudes of herbivory compared to later genotypes (Choi et al., 2019; Polgar and Primack, 2011). Similarly, genotypes with shorter budbreak and leaf expansion durations might be exposed to less herbivory compared to genotypes with longer durations, which has been shown in comparisons of the duration of leaf expansion across species (Kursar and Coley, 1991; Moles and Westoby, 2000). The intraspecific phenological variation in our study might help to explain the effects of genotype on previously documented variation in herbivore community composition on *S. hookeriana*, where variation in leaf chemistry has been shown to be a primary predictor of variation in the herbivore community observed on these plants (Barbour et al., 2015, 2016). In terms of reproduction,

genotypes that flower earlier might experience more or less effective pollination compared to later-flowering genotypes. Given that *S. hookeriana* is one of the earliest floral resources in the spring in these coastal communities (M. Mesler, Cal Poly Humboldt, personal communication), early flowering might be beneficial if flowering later results in increased competition for pollinators as other species begin to flower (sensu Augspurger, 1983; but see Waser, 1978). The mid-flowering genotypes could have a fitness advantage if a threshold number of conspecific flowers is necessary for attracting pollinators to the population (Heithaus et al., 1982). Conversely, mid- and/or later-flowering genotypes might risk satiating pollinator activity and experience reduced fitness (Zimmerman, 1980). All these hypotheses assume that pollinator visitation is related to seed production, which is certainly not always the case, nor is seed production necessarily a good indicator of recruitment and population growth (Price et al., 2008; Lundgren et al., 2015). Less is known about the advantages and disadvantages of shorter vs. longer flowering duration compared to flowering date (Li et al., 2021), but shorter flowering durations could increase reproductive fitness if this results in less heterospecific pollen transfer. The trade-off is that shorter durations might also result in a more restrained time for pollination, with less buffering capacity for periods of low pollinator availability or environmental uncertainty compared to longer flowering durations (Rathcke and Lacey, 1985). These questions have been investigated for the longevity of individual flowers (Primack, 1985) but much less so for the duration of flowering at the individual level.

CONCLUSIONS

In conclusion, we demonstrated that genetic variation has clear effects on the timing and duration of multiple, sequential life-cycle events in *S. hookeriana*, explaining on average more than half (62.3%) of the within-population variation in timing and duration. We also documented a high degree of variation across individuals in the timing and duration of events, comparable to levels of interspecific variation in phenological traits. The duration of life-cycle events has received far less study than the timing of events, but given the comparable amount of intraspecific variation in event durations and in timing and that duration is important for fitness, ecosystem processes, and ecosystem services, the degree to which genes and environment affect the duration of life-cycle events should receive further study.

AUTHOR CONTRIBUTIONS

E.T.W. and E.S.J. conceived of the study; E.S.J., E.T.W., P.J.C., G.A.F., K.N.G., A.J.G., K.C.P., N.W., M.D.W., G.A.E., and C.S. conducted fieldwork; L.K.R., A.M.I., and P.J.C. processed and cleaned data; A.M.I. and P.J.C. analyzed data; A.M.I. wrote the paper; all authors contributed edits.

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DATA AVAILABILITY STATEMENT

The data and R script required to replicate all analyses in this paper are archived in the Open Science Framework account of Amy M. Iler under the project name “Willow Phenology”: <https://doi.org/10.17605/OSF.IO/N6GS2>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. Photographs of the common garden.

Appendix S2. Model selection tables for timing (Table S1) and duration (Table S2) of life-cycle events.

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